

How to Replicate Your Dragonfruit: A replication of single-cell and spatial RNA findings on fruit senescence

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Background

- Li et al. (1) uses scRNA-seq and ST to examine the changes in transcription in different cell types during senescence.
- Fruit senescence creates lost revenue and food waste. Understanding this process could help identify targets for improving shelf life.
- The original paper sought to identify cell types, expression patterns and spatial patterns associated with senescence.
- This study aimed to determine whether similar cluster structure, marker genes, and senescence-related patterns can be found with modified methods and the information provided.

Methods

A bioinformatics pipeline was constructed that followed the original methods as closely as possible. Python equivalents were substituted in the analysis step (Figure 1). As the Visium image was not available, the published cell atlas information was used in lieu of manual integration.

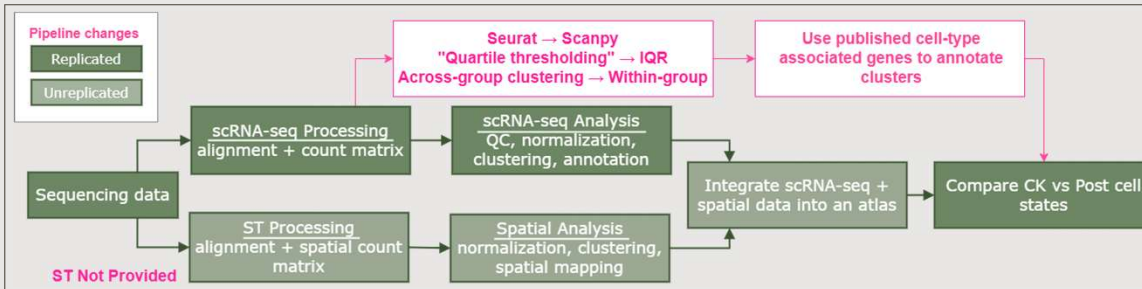


Figure 1. The pipeline of the replicated paper, with changes highlighted.

Results

Study challenges and reproducibility

- Pipeline was implemented for the pitaya scRNA-seq data, but spatial integration was impossible due to unavailable data.
- A few minor assumptions were required, including how quartile thresholding was implemented.
- Without biological replicates, batch effects could not be removed.
- There were no identifiable mitochondrial genes in the pitaya reference, limiting mitochondrial-based QC.

Cell analysis

- The senescent and mature samples showed strong separation of RNA expression. This agrees with the original paper's conclusion that senescence is associated with major cell-state changes.
- The condition split in the replication was stronger than in the published clustering of scRNA, suggesting that the approximation did not fully reproduce the study structure.
- There was a clear decrease in exocarp and increase in mesocarp from mature to senescent samples

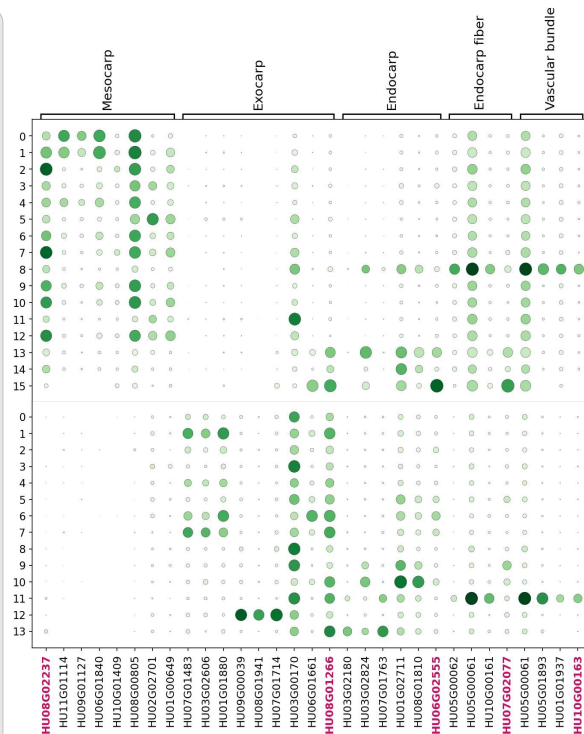


Figure 2. Dotplot of the senescent (top) and mature (bottom) sample clusters.

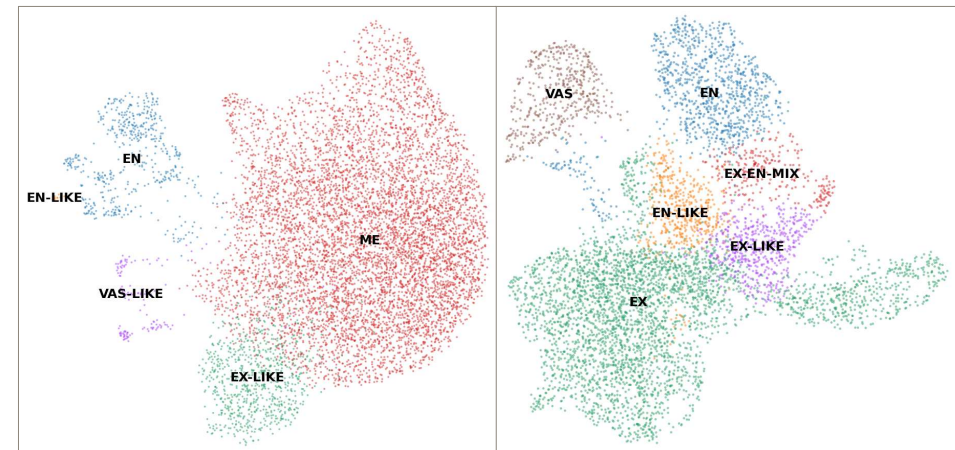


Figure 3. UMAP of cells, coloured by cell type for the senescent (left) and mature (right) samples. EX=Exocarp, EN = Endocarp, ME = Mesocarp, VAS = Vascular bundle.

Conclusions

The replication partially reproduced the original study. Broad cell-type marker patterns and the mature to senescent shift were recovered, supporting the idea that the original findings reflect real biological structure. The discrepancy in the clustering split suggests the missing details and data limited reproducibility.

Future Directions

- Re-run the analysis with spatial mapping to test improvements to clustering annotation.
- Redo the sampling steps, with biological replicates to rule out batch effects.
- Rebuild the reference with mitochondrial annotation or obtain a reference that includes organellar genes to conduct better QC.
- Perform pseudotime analysis of EX and ME.

Table 1. Comparison of original outputs with the current study.

Original	Current
5645 Mature and 7670 Senescent cells after filtering	6331 Mature and 8612 Senescent cells after filtering
Broad cell types narrowed with ST	Broad cell type groups
Shared mature and senescent atlas structure	Stronger separation by condition
Mature sample: Exocarp-enriched.	Mature sample: Exocarp-enriched
Senescent sample: Mesocarp-enriched.	Senescent sample: Mesocarp-enriched.